ADITYA UNIVERSITY

Department of freshman Engineering

Important Questions

Course name: Data Structures **Course code:** 2501CS02

UNIT-1					
QNO	**Sub QNO**	**Question**	**BT**	**CO**	**Marks**
1	a	Define structure. Explain creation, accessing and initialization of structure with suitable examples.	L2	CO1	4M
	b	Write a C program to create a structure named Student with roll number, name, and marks. Read and display the details.	L3	CO1	4M
2	a	Illustrate the concept of nested structure with examples	L2	CO1	4M
	b	Write a C program using typedef to define a new data type for structure.	L3	CO1	4M
3	a	When array of structures needed. Justify your answer with an example	L2	CO1	4M
	b	Write a C program to calculate total and average marks of a student using functions and structure.	L3	CO1	4M
4	a	Distinguish array, structure and union with examples	L2	CO1	4M
	b	Write a C program to declare an enum for days of the week and print any given day.	L3	CO1	4M
5	a	Describe self referential structures with an example.	L1	CO1	4M
	b	Write a C program to store and display details of 5 students using an array of structures.	L3	CO1	4M
6	a	Write a short notes on following terms i) Enum ii) typedef iii) Bit fields	L2	CO1	4M
	b	Program	L3	CO1	4M
7	a	Explain the classification of data structures with suitable examples	L2	CO1	4M
	b	Write a C program to define a structure with bit fields.	L3	CO1	4M
8	a	Demonstrate the concept of structures and functions	L2	CO1	4M
	b	Write a C program to store and display details of a book (title, author, price) using structure.	L3	CO1	4M
9	a	Define union. Explain with suitable examples	L2	CO1	4M
	b	Write a C program to define a structure Employee that contains another structure Address (city and pincode). Read and display employee details.	L3	CO1	4M
UNIT-2					
QNO	**Sub QNO**	**Question**	**BT**	**CO**	**Marks**
1	a	Define algorithm. Explain the process to find the efficiency of an algorithm with suitable examples	L2	CO2	4M

	b	Describe with suitable example about quick sort algorithm(procedure, algorithm, program)	L3	CO2	4M
2	a	Explain Asymptotic notations with necessary graphs.	L2	CO2	4M
	b	Describe with suitable example about merge sort algorithm(procedure, algorithm, program)	L3	CO2	4M
3	a	Define ADT. Explain different type ADTs with their operations.	L2	CO2	4M
	b	Describe with suitable example about Radix sort algorithm(procedure, algorithm, program)	L3	CO2	4M